EXERCISE 06

Poles and Zeros

RECAP

Where we left off...

- Last week we only needed one transformation pair. For an input $u(t) = e^{at}$:

$$\mathcal{L}\{e^{at}\} = _____ = U(s)$$

$\Rightarrow U(s)$ has a _____ at $s = a$: it tells us which exponential the signal is made of.

$\Rightarrow$ For $s \neq a$, $U(s)$ is not 0: another exponential e^{st} is not totally different from e^{at}, it is still _____ (both grow or decay, just at a different rate).

- Every SISO-LTI-system can be written as a fraction of two polynomials in frequency domain.

Time domain

$$\dot{x}(t) = Ax(t) + Bu(t)$$
$$y(t) = Cx(t) + Du(t)$$

Laplace-Transform

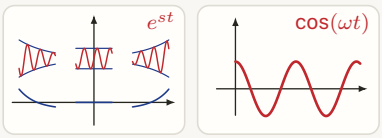
$$G(s) = C(s\mathbb{I} - A)^{-1}B + D$$

Frequency domain

$$G(s) = \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \cdots + b_0}{s^n + a_{n-1}s^{n-1} + \cdots + a_0} + d$$

- The variable s is a “complex frequency”: $s =$ _____, where
  - $\mathrm{Re}(s) = \sigma$: _____ part
  - $\mathrm{Im}(s) = \omega$: _____ part

$\Rightarrow$ In this course, most of the time we have _____ (pure sinusoids).



1 | Sinusoids through a transfer function

Input signals as exponentials

- We saw, that most signals can be written as

$u(t) =$ ..... $\Rightarrow$ EX. $\cos(\omega t) =$ .....

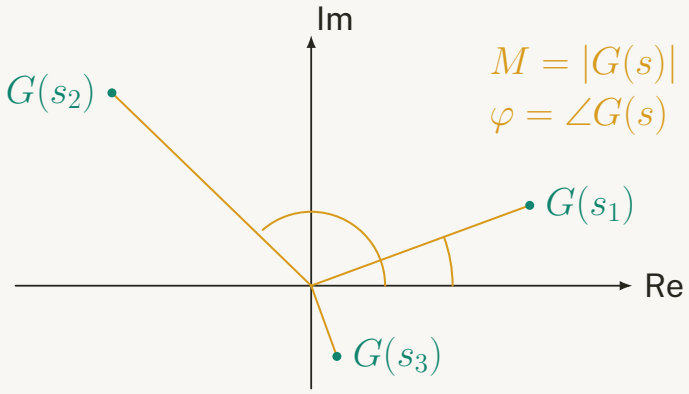
- Since we had that $u \longrightarrow \boxed{G(s)} \longrightarrow y$, we find that

$$y_{ss}(t) = \text{.....}$$

but if we evaluate this for some $s = \sigma + j\omega$, we get weird complex values...

$\Rightarrow$ $y_{ss}(t) =$ .....

$\Rightarrow$ Plotting this for different s we can find:

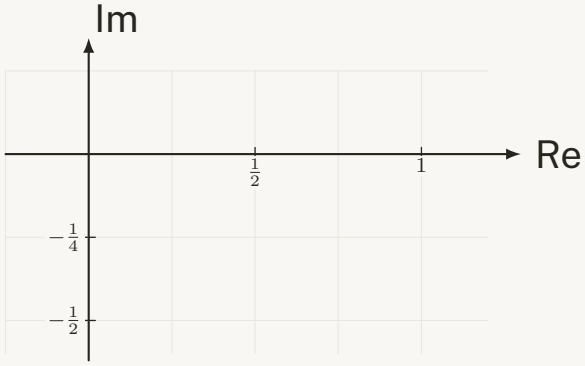


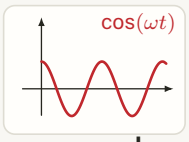
$M = |G(s)|$: Magnitude
$\varphi = \angle G(s)$: Phase

$$a + jb = M\, e^{j\varphi}$$

EX. Take $G(s) = \dfrac{1}{s+1}$ and plug in pure sinusoids $s = j\omega$:

s	$G(s)$	M	φ
0	.....	.....	.....
j	.....	.....	.....
$2j$	.....	.....	.....





$\Rightarrow G(s)$ evaluated at some s is just a complex number: a point in the complex plane with magnitude M and phase φ. Here: the higher the frequency ω, the _____ the amplitude and the _____ the phase lag.

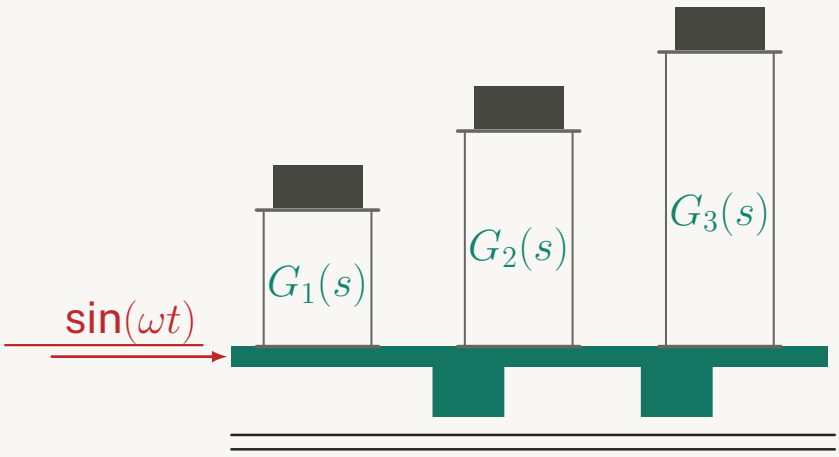
$\Rightarrow$ We can write

$$y_{ss}(t) = \tfrac{1}{2}\, M e^{j\varphi}\, e^{j\omega t} + \tfrac{1}{2}\, M e^{-j\varphi}\, e^{-j\omega t}$$

$$= M \cdot \tfrac{1}{2}\left(e^{j(\omega t+\varphi)} + e^{-j(\omega t+\varphi)}\right)$$

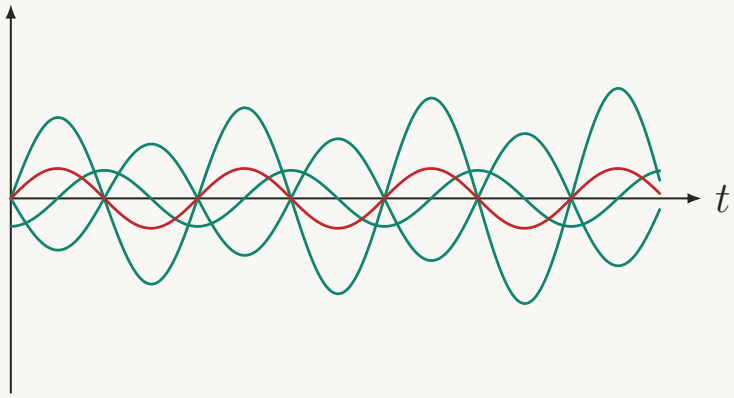
$\Rightarrow$ which is just $y_{ss}(t) =$ _____ !!!

INTUITION

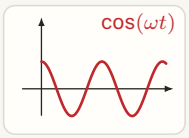


https://www.youtube.com/watch?v=LV_UuzEznHs

$\Rightarrow$ For the same s, these systems have different responses.



- Inputs experience a change in



- Amplitude: $|G(s_i)|$
- Phase: $\angle G(s_i)$,

depending on input frequency.

⇒ Output frequency does not change!!

Is that the whole story?

Now we have fully characterized systems in frequency domain, right? No...

- Take another look at the formula for the transfer function $G(s)$...

$$G(s) = C(s\mathbb{I} - A)^{-1}B + D$$

⇒ Notice how we said a LTI-system is fully characterized by the following equation:

$$y(t) = \overbrace{Ce^{At}\,x(0)} + \overbrace{\int_0^t Ce^{A(t-\tau)}\,B\,u(\tau)\,d\tau \;+\; \underline{D}\,u(t)}$$

⇒ All parameters included in TF formula, except: _____ !!

⇒ TF's only show the _____ response!

So far so good? Let's dive in-depth into TF-structure.

2 | Poles and zeros

Transfer functions: forms

- STANDARD FORM

$$G(s) = \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \cdots + b_0}{s^n + a_{n-1}s^{n-1} + \cdots + a_0} + d$$

- ROOT-LOCUS FORM

$$G(s) = \frac{k_H}{s^q} \cdot \frac{(s - z_1)(s - z_2)\cdots(s - z_m)}{(s - p_1)(s - p_2)\cdots(s - p_{n-q})}$$

- BODE FORM

$$G(s) = \frac{k_{Bode}}{s^q} \cdot \frac{\left(\frac{s}{-z_1} + 1\right)\left(\frac{s}{-z_2} + 1\right)\cdots\left(\frac{s}{-z_m} + 1\right)}{\left(\frac{s}{-p_1} + 1\right)\left(\frac{s}{-p_2} + 1\right)\cdots\left(\frac{s}{-p_{n-q}} + 1\right)}$$

- The scalar values _____ are called _____ of $G(s)$
- The scalar values _____ are called _____ of $G(s)$

DEF Poles $\Rightarrow$ roots of the denominator of $G(s)$

Zeros $\Rightarrow$ roots of the numerator of $G(s)$

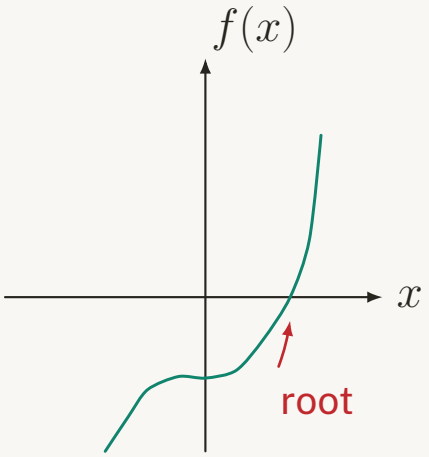
$$G(s) = \frac{N(s)}{D(s)} \quad \Rightarrow \quad \begin{aligned} z_i &: ____ \text{ s.t. } ____ \\ p_i &: ____ \text{ s.t. } ____ \end{aligned}$$

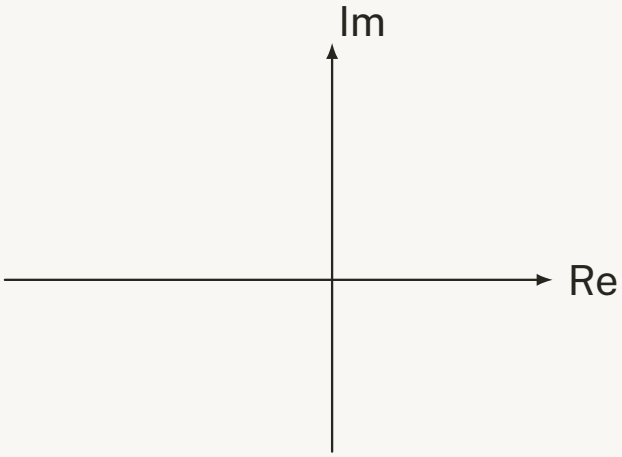
- Fundamental theorem of algebra:

_____ of polynomial $\widehat{=}$ _____ (complex) roots

Example: roots of a polynomial

EX. $f(x) = x^3 + x^2 - 2 = 0 \Rightarrow$ .....





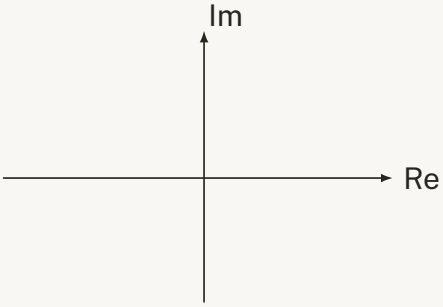
$\Rightarrow$ The location of poles & zeros are the defining characteristic of an LTI-system.

- Graphical notation (given root-locus form)

$$G(s) = \frac{k\,(s+2)(s-1)}{s\,(s+1+j)(s+1-j)}$$

$\times$: poles, $\circ$: zeros

$\rightarrow$



Poles

RECALL $G(s) = C(s\mathbb{I} - A)^{-1}B + D$

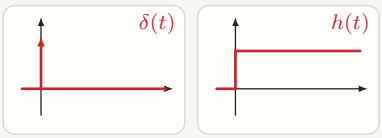
From linear algebra, we had that $A^{-1} =$ .....

$$\Rightarrow\ G(s) = \frac{1}{\det(s\mathbb{I} - A)} \cdot \overbrace{\left(C\,\mathrm{adj}(s\mathbb{I} - A)B + D\det(s\mathbb{I} - A)\right)}^{\text{.....}} = \frac{N(s)}{\det(s\mathbb{I} - A)}$$

$\Rightarrow$ Setting $\det(s\mathbb{I} - A) = 0$ is how we find ..... !

..... of $G(s) \mathrel{\widehat{=}}$ Eigenvalues of .....

Position of poles will tell us about ....., ... behaviour



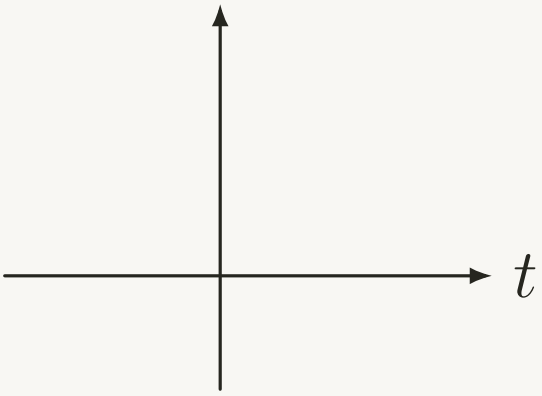
and _____ behavior!

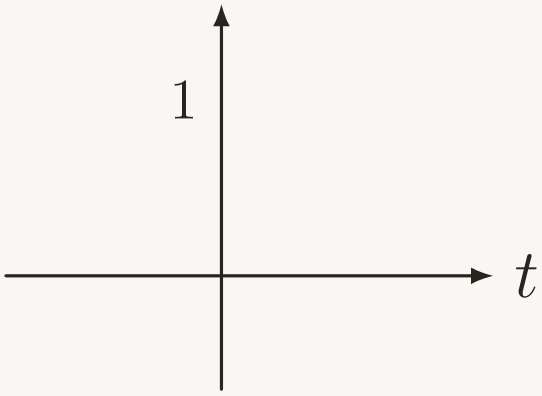
Test signals

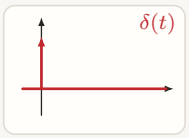
- To visualize the effect of the position of poles and zeros, we will use two fundamental input signals: $\Rightarrow$ _____-signals

TEST-SIGNALS (Some examples)

- _____: $\delta(t) = \begin{cases} _____ \\ _____ \end{cases}$
- _____ function: $h(t) = \begin{cases} _____ \\ _____ \end{cases}$ (_____ function)







3 | Effect of poles: the impulse response

$\Rightarrow$ Why are they relevant? Let's look at $\delta(t)$:

- The reaction of a system to a $\delta(t)$ input is called *Impulse-response*:

$$Y(s) = G(s)U(s) = \text{.....}$$

$$\Rightarrow\ Y(s) = G(s) \quad | \ \text{.....}$$

$$\Rightarrow\ y(t) = \underbrace{\mathcal{L}^{-1}\{G(s)\}}$$

How do we find this?

$x(t)$	$X(s)$
impulse: $\delta(t)$	1
step: $h(t)$	$\frac{1}{s}$
$h(t) \cdot e^{at}$	$\frac{1}{s-a}$
$h(t) \cdot \sin(\omega t)$	$\frac{\omega}{s^2+\omega^2}$
$h(t) \cdot \cos(\omega t)$	$\frac{s}{s^2+\omega^2}$

- ..... expansion

$$G(s) = \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \cdots + b_0}{s^n + a_{n-1}s^{n-1} + \cdots + a_0}$$

$$G(s) = \underbrace{\frac{r_1}{s-p_1}} + \underbrace{\frac{r_2}{s-p_2}} + \cdots + \underbrace{\frac{r_n}{s-p_n}}$$

r_i: .....

Of these terms we can easily find $\mathcal{L}^{-1}\{\cdot\}$

$$\Rightarrow\ y(t) = \mathcal{L}^{-1}\left\{\frac{r_1}{s-p_1} + \frac{r_2}{s-p_2} + \cdots + \frac{r_n}{s-p_n}\right\}$$

| Laplace is ..... !

$$\Rightarrow\ y(t) = \text{.....}$$

Now we can think of the effect of different poles.

RECALL $s = \sigma + j\omega$

- $e^{\sigma t}$, $\sigma < 0$



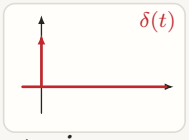
- $e^{\sigma t}$, $\sigma > 0$



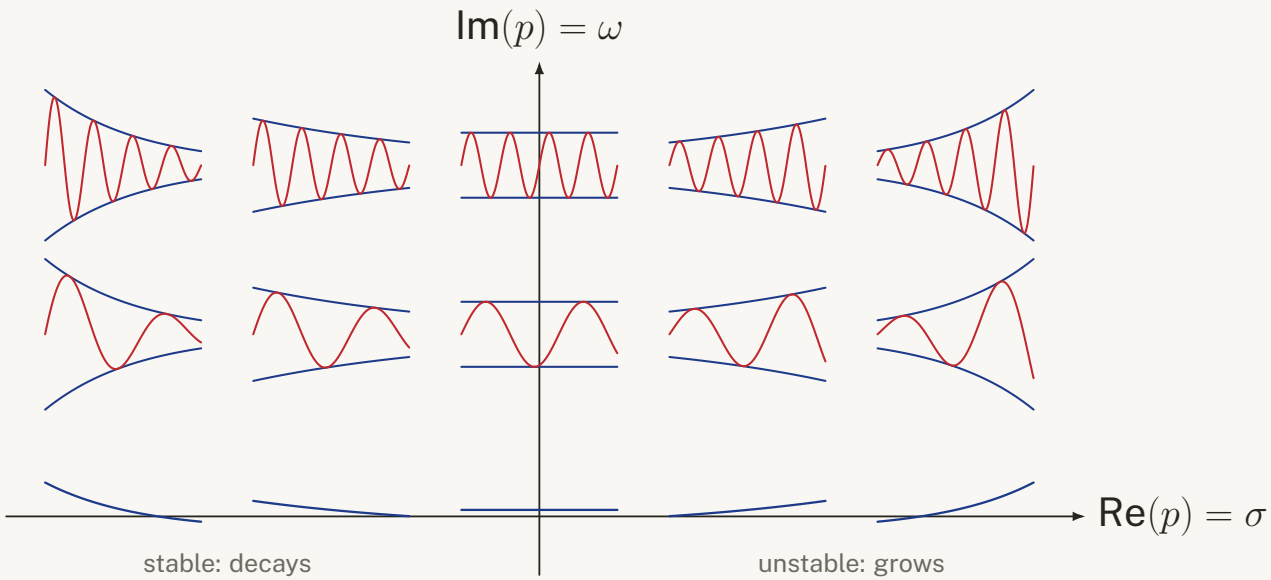
- $e^{j\omega t} \pm e^{-j\omega t}$



$\Rightarrow$ Matlab

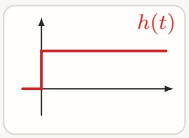


OVERVIEW Shape of the mode e^{pt} for different pole positions $p = \sigma + j\omega$:



$$Y(s) = \underbrace{G(s)}_{\text{system}} \cdot \underbrace{U(s)}_{\text{input}}$$

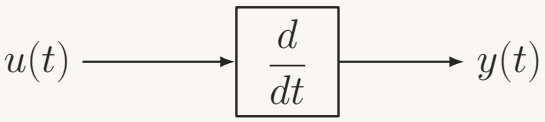
$\Rightarrow$ The poles p in this figure are the poles of the system $G(s)$: they fix its modes e^{pt}, no matter which input we apply.



4 | Effect of zeros

Zeros and derivatives

To understand the effect of zeros, consider this:



$$y(t) = \frac{d}{dt}u(t)$$
$$\Downarrow$$
$$Y(s) = \text{.....}$$

RECALL THIS PROPERTY:

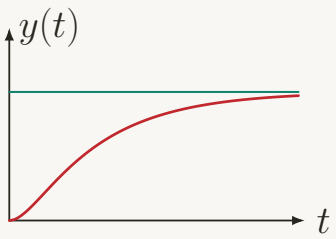
$$\mathcal{L}\{f^{(n)}(t)\} = s^n F(s)$$

New ..... in TF!

$\Rightarrow$ Zeros have some connection to derivatives...

How does adding zeros to a system affect their responses?

- Let's try and see this with step responses
- Assume some system $G(s)$ without zeros and as input, apply a step $h(t)$:



$\Rightarrow$ How would this change if we added a zero?

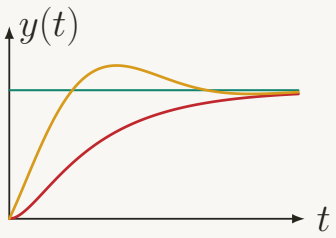
$\tilde{G}(s) =$ .....

$=$ .....

$\Rightarrow$ The "s" acts as a derivative so the output would look like:

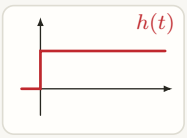
$\tilde{y}(t) =$ .....

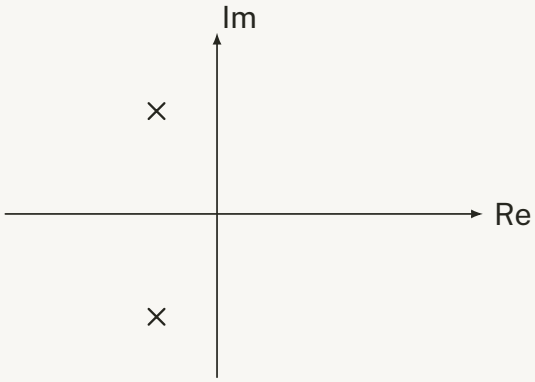
We add on top a ..... derivative of $y(t)$!



The zero has ..... effect! It "knows in advance" that it is off from the goal.

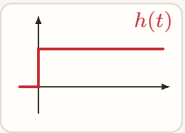
- Placement of zeros







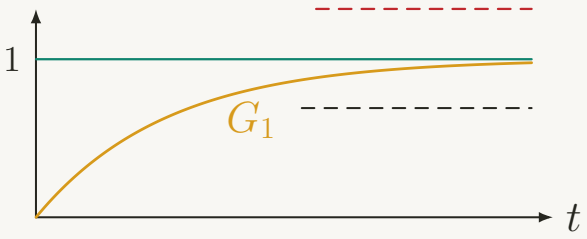
- Zeros in left-half plane: zeros
- Zeros in right-half plane: -minimumphase zeros
  - Think of how the location of z affects $\tilde{y}(t) = -\frac{1}{z}\dot{y}(t) + y(t)$



5 | Steady state: the final value theorem

- The step response is a really useful tool to analyze systems.

$\Rightarrow$ One thing that we have glossed over...



- A system doesn't necessarily converge to the input value!

$$y_{ss}(t) = \quad \text{.....}$$

- So how could we quantify this ..... ?

..... Theorem

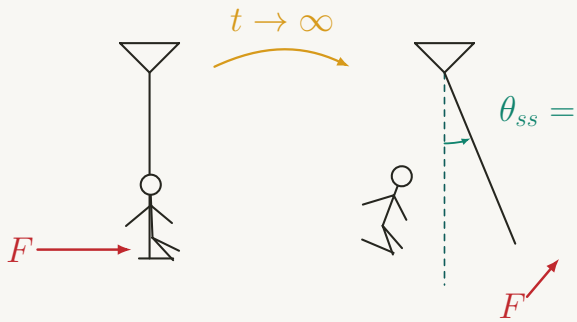
$$\lim_{t\to\infty} y(t) = \quad \text{.....} \quad = \quad \text{.....}$$

$x(t)$	$X(s)$
impulse: $\delta(t)$	1
step: $h(t)$	$\frac{1}{s}$
$h(t) \cdot e^{at}$	$\frac{1}{s-a}$
$h(t) \cdot \sin(\omega t)$	$\frac{\omega}{s^2+\omega^2}$
$h(t) \cdot \cos(\omega t)$	$\frac{s}{s^2+\omega^2}$

- Step function: $y_{ss} =$ .....

$$y_{ss} = \lim_{s\to 0} \quad \text{.....} \quad = \lim_{s\to 0} \quad \text{.....}$$

Example: step force on a swing

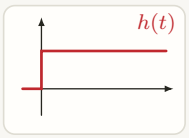


$$\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ \frac{g}{\ell} & -\frac{c}{m\ell^2} \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ \frac{1}{m\ell} \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} x(t)$$

$$F(t) = \begin{cases} 0, & t < 0 \\ 1, & \text{else} \end{cases}$$

- We could now solve this as



$$\theta_{ss} = \lim_{t\to\infty}\left(Ce^{At}x_0 + \int_0^t Ce^{A(t-\tau)}B\,d\tau\right)$$

Good luck...

Better idea: transform to frequency domain!

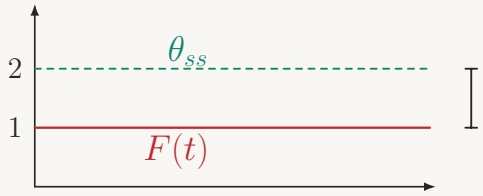
$$G(s) = C(s\mathbb{I} - A)^{-1}B + D = \frac{4}{s^2+s+2}$$ | Random params

Then use final value theorem

$\theta_{ss} = \lim_{s\to 0} s\,G(s)\,U(s)$, $U(s) = H(s) = \frac{1}{s}$ (step-function)

$= \lim_{s\to 0} \frac{4}{s^2+s+2} =$ .....

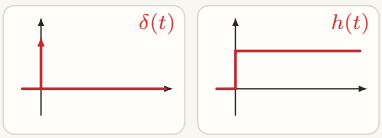
(Degrees, radians,...)



This factor has a name...

$\Rightarrow\ G(s\to 0) =$ .....

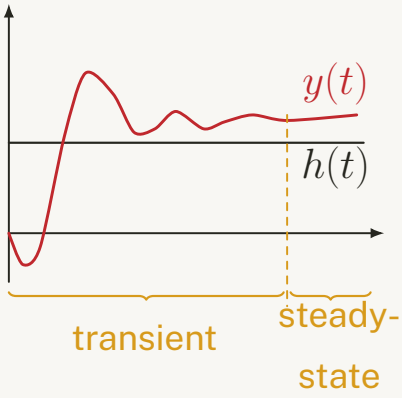
“ ..... ”



6 | Summary and a warning

In summary

$\Rightarrow$ If you have a system $G(s)$ and want to know how it responds to a step/impulse



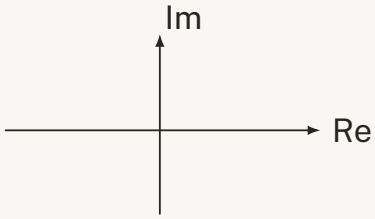
Transient: .....

Steady-state: .....

Pole-zero cancellation

- Consider a system with a zero near a pole

$$G(s) = \frac{s+1+\varepsilon}{(s+1)(s+2)}$$



what happens if ..... ?

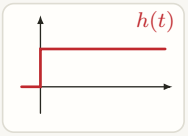
$$G(s) = \frac{s+1}{(s+1)(s+2)}$$

The pole and the zero seem to cancel!

But they dont!!!

The internal dynamics remain, we just don't see them anymore in the output!

Dangerous if ..... .



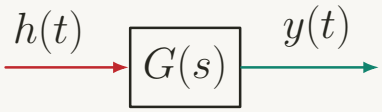
Integrators and differentiators

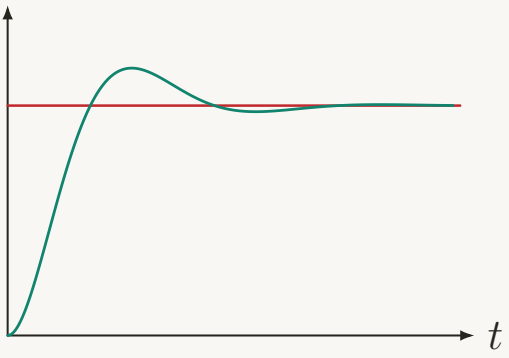
Poles at $p = 0$: Integrators

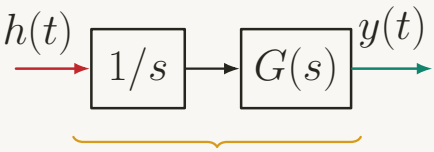
Zeros at $z = 0$: Differentiators

$\Rightarrow$ We will use this in controller design

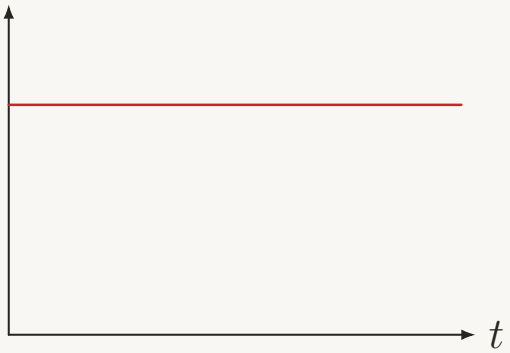
$$G(s) = \frac{1}{s^2 + s + 1}$$

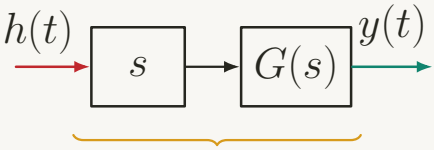




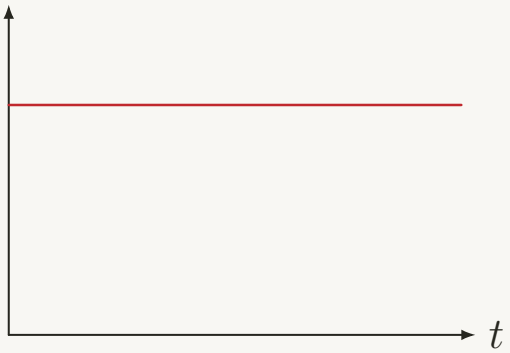


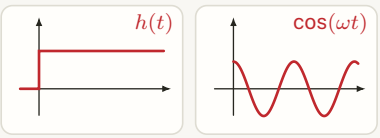
$$\frac{1}{s^3 + s^2 + s}$$





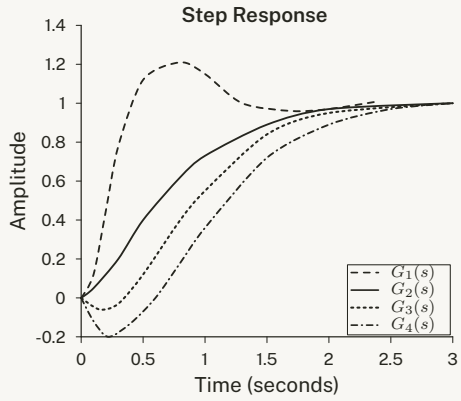
$$\frac{s}{s^2 + s + 1}$$

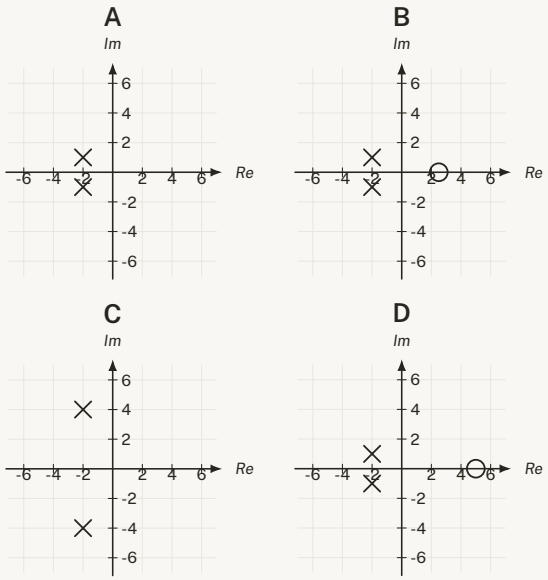




QUIZ

Q1) Match poles/zeros to step response





Q2)

Problem: Consider a linear time-invariant system with transfer function $G(s)$ that is given by a root-locus gain $k_{rl} = 1$ and by two poles and two zeros as shown in Figure 7.

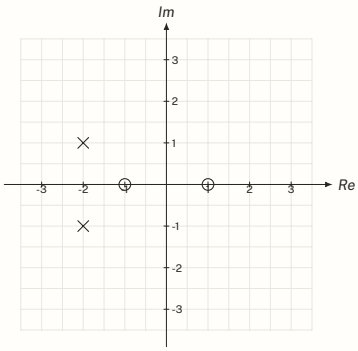


Figure 7: Poles ($\times$) and zeros ($\circ$) of the LTI system $G(s)$.

Q19 (1 Points) Mark the correct answer.

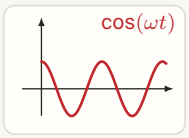
What is the magnitude $|G(j)|$ of $G(s = j)$?

A $|G(j)| = \frac{1}{2\sqrt{2}}$ B $|G(j)| = \frac{1}{4}$ C $|G(j)| = \frac{1}{2}$ D $|G(j)| = \frac{1}{\sqrt{2}}$

Q20 (1 Points) Mark the correct answer.

What is the phase $\angle G(j)$ of $G(s = j)$?

A $\angle G(j) = 45°$ B $\angle G(j) = 0°$ C $\angle G(j) = -45°$ D $\angle G(j) = 135°$



SOLUTION

Poles: .....

Zeros: .....

$G(s) =$ .....

$$G(\text{.....}) = \frac{(j-1)(j+1)}{2(2+2j)} = \frac{-2}{4+4j} = -\frac{1}{2+2j}$$

- $|G(s=j)| = \dfrac{|-1|}{|2+2j|} =$ ..... $=$ ..... $=$ ..... //

- $\angle G(s=j) = \angle\dfrac{-1}{2+2j} =$ .....

  $=$ ..... $=$ .....

  $=$ ..... $=$ ..... $=$ ..... //